[PLACEHOLDER TITLE]

Starting from the past: transient initialization and the input-history problem in soil carbon models

Lorenzo Menichetti [co-authors TBD]

Draft outline — July 15, 2026

Abstract

[PLACEHOLDER ABSTRACT.] One paragraph: steady-state initialization is a near-universal but rarely examined convention; an equilibrium start is already at its saturated end-state, so it cannot reproduce an accumulating sink that is only now beginning to saturate — the very dynamic observed in Finnish forest soils. We test a transient alternative across six models under one fair frame; it captures the accumulation-then-saturation as a proof of concept, but relocates the binding uncertainty onto litter inputs — especially their unknown history — pointing to historical, interdisciplinary reconstruction as the real frontier.

Contents

Part I
Opening

1 Introduction

1.1 The convention: steady state at t_0

A near-universal default in SOC modelling. State it plainly as common practice, not yet as a problem.

1.1.1 What the assumption actually claims

That the system's past equals its present equilibrium (inputs $\approx$ outputs at the start). Make the hidden content of the convention explicit.

1.1.1.1 Why it is attractive

Convenient, analytically clean, needs no historical forcing. The reasons it became default.

1.1.1.2 Why it is rarely examined

It is invisible precisely because it is standard. Set up the reader to recognize their own practice.

1.1.2 The non-equilibrium reality

Real ecosystems are transient; they carry the imprint of disturbance, land-use and productivity history.

1.1.2.1 An equilibrium start is already at the end

A steady-state start sits at zero net change — by construction already at its saturated end-state. It forecloses the accumulation phase before the simulation begins. Foreshadow the capture-failure without resolving it.

1.1.3 Managed landscapes are rarely at equilibrium

Land-use and management histories keep soils in transient states; equilibrium is the exception, not the rule. Flag Finnish forest history as the concrete case that justifies the violated assumption — stated generally here, examined in detail in the Discussion.

1.2 Why this matters

1.2.1 Scientific stakes

Initialization choices propagate into every downstream projection and into model intercomparison.

1.2.2 Policy stakes: the motivating case

Finland reports its forest soil as a sink; climate policy leans on it. Introduce Finland as the instance, not the subject.

1.2.2.1 The observed dynamic

Accumulation (VMI8→Biosoil), then recent hints of saturation (Komeetta, under review). A sink caught in the act of saturating. State the phenomenon the model must reproduce.

1.3 Scope and contribution (preview)

One short paragraph previewing the move: a transient initialization, tested fairly across a model ladder. Keep it a promise, not the argument.

Part II
Challenge

2 The gap and the question

2.1 The saturation an equilibrium start cannot show

The triggering insight: to reproduce a sink that is saturating, the model must first reproduce the accumulation it is the tail of — and an equilibrium start, already saturated, cannot. This is what motivated transient init.

2.1.1 Why this generalizes beyond Finland

Any non-equilibrium system started at steady state forecloses its own transient. Lift the problem off the case study.

2.1.2 Artefact or structure?

Do divergent saturation projections reflect genuine model structure, or the initialization and calibration convention? The question the fair frame will answer.

2.2 The methodological gap

No established, calibratable way to initialize a transient run when historical forcing data are absent.

2.2.1 The fixed-value trap

Treating the unknown initial state as a single fixed number (equilibrium, or a guessed value) hides a strong, untested assumption inside the result.

2.2.2 Calibrate it, do not assume it

The alternative: make the historical start a parameter with a prior and let the data and its uncertainty speak. Plant the thread; pay it off in Methods.

2.3 Objectives and hypotheses

- H1 Transient initialization lets the models reproduce the accumulation-then-saturation a steady-state start cannot. [placeholder]
- H2 Residual model differences under a fair frame are structural and modest. [placeholder]
- H3 The binding uncertainty migrates to inputs, not kinetics. [placeholder]

Part III
Action

3 Materials and methods

3.1 Study system and data

3.1.1 Plots and SOC campaigns

Finnish NFI plots; three temporally separated SOC campaigns as targets. Brief.

3.1.2 Litter inputs

Plot-level AWEN-fractionated inputs; note the population-mean rise then plateau (load-bearing later).

3.1.3 Climate forcing

Gridded daily climate; per-model temporal aggregation.

3.2 The model ladder

Six models on two crossed axes (pool complexity; climate integration), so no conclusion rests on one structure.

3.2.1 Pool-complexity axis

SP1 → TP2 → TP3 → Yasso07. One sentence each. [placeholder]

3.2.2 Climate-integration axis

Yasso07 → Yasso15 → Yasso20. One sentence each. [placeholder]

3.3 Bayesian calibration: the enabling machinery

The inference framework that lets the unknown start be calibrated rather than assumed. Keep it brief and matter-of-fact; its role is enabling, not headline.

3.3.1 Why a Bayesian frame is the natural fit

Priors encode what little is known about the past; the posterior carries uncertainty forward instead of collapsing it to a point.

3.3.2 Uncertainty that propagates, not a likelihood patch

σ_{init} is a prior on the historical state; its spread flows into both the initial condition and the forward dynamics. Distinguish from a mere error term added at the end.

3.4 Transient initialization (the protagonist)

3.4.1 Pushing the equilibrium assumption back in time

Initialize at analytical steady state in the past, run transiently forward to t_0 so the imprint dissipates and the model arrives at t_0 still accumulating.

3.4.1.1 The fast/slow relaxation argument

Fast pools fully relax; only the slow humus pool retains memory — it "remembers" the equilibrium start. Why the pre-run works, and where its memory lives.

3.4.2 Calibrating the historical anchor

σ_{init} as a Bayesian-calibrated strength on the historical start, not a fixed choice — the concrete place the thread meets the protagonist. Full pre-run construction in Appendix C.

3.5 A fair frame for intercomparison

The design principle: any residual difference must be attributable to structure alone.

3.5.1 Shared data, likelihood, forcing, filtering

Brief enumeration.

3.5.2 Exact integration in every model

All six integrate their linear system exactly; differences are structural, not numerical. Solvers and the discarded Euler path in Appendix E.

3.5.3 Homogenized priors: method, not number

Tiered scheme; identical in construction/scale/DoF, not in raw values. Full parameter-class × prior-criteria tables in Appendix B.

3.5.4 Degrees of freedom, set by external information

Flow fractions are equal-DoF (all free) in every multi-pool model — the fair-frame core. Rates differ BY DESIGN, and the reason is information, not inconsistency: Pin what external knowledge lets you pin; calibrate the rest. [strand] This convention is also what makes the stratified extension Yasso-only: with rates fixed, Yasso's stratifiable knobs are inputs + the humus-formation fraction — see the optional strand in the Outlook.

3.6 Calibration and inference

3.6.1 Error model

Multiplicative-normal likelihood. Flag the asymmetry briefly (paid back in Discussion).

3.6.2 Physically-bounded input priors

The effective litter flux is bounded to a boreal NPP envelope, homogeneous across models, so the input multiplier cannot manufacture unphysical carbon; the auxiliary $\sigma_{\text{init}}/\sigma_{\text{input}}$ pair is re-parameterised rather than truncated. Rationale, envelope and mechanism in Appendix A. [placeholder]

3.6.3 Sampler and diagnostics

MCMC settings; convergence and identifiability diagnostics used.

3.6.4 Holdout split and validation

Plot-level holdout excluded from the likelihood; separate predictive metrics. [placeholder]

3.6.5 Software, compute and reproducibility

Engine, samplers, exact-integration back-ends; fixed seeds; HPC production run. [placeholder]

3.7 Predictive and residual analysis

Posterior-predictive trajectories; residual analysis against site covariates.

4 Results

4.1 Capturing accumulation-then-saturation

The headline: transiently-initialized models reproduce the rise and its incipient bend-over; the equilibrium-started flat/declining shape was the start, not the structure.

4.1.1 Trajectory shape vs. level

Shape change, not just offset. [placeholder figure]

4.1.2 Match to campaign means

2006 and 2024 means tracked. [placeholder]

4.2 Structural differences under the fair frame

Real but modest skill spread across the six. [placeholder table]

4.2.1 Where simple beats complex

TP2/TP3 track shape best; stiffer Yasso family worst. Brief.

4.3 What the data can and cannot constrain

Set up the Discussion: kinetics weakly constrained; inputs do the work.

4.3.1 Convergence vs. identifiability

Good R-hat coexisting with non-identified fractions. State the apparent paradox, resolve in Discussion.

4.3.2 Input multiplier behaviour

σ_{input} does different work in different models. The hinge result.

4.4 Residual structure

RF importance; stand basal area as the leading, diffuse, shared driver. [placeholder figure]

Part IV
Resolution

5 Discussion

5.1 Capturing saturation requires remembering the accumulation

Resolve the Challenge: the difficulty was the equilibrium start, not the soil; a transient start lets every model represent the observed dynamic. The central payoff.

5.1.1 Proof of concept, not a verdict on Finland

What the result does and does not license. Calibrate the claim.

5.1.2 Generalization to the model class

Wherever steady-state init meets a non-equilibrium system, it forecloses the transient the data demand. Lift off Finland.

5.2 The binding uncertainty is the inputs (the σ_{input} reveal)

The data cannot resolve kinetics; the input multiplier absorbs what structure cannot. So the frontier is inputs, not soil chemistry.

5.2.1 The multiplier crosses a physical line (quantitative payoff)

The physical bound turns this into a clean counterfactual, and that counterfactual is the cleanest proof that inputs — not kinetics — were the lever. Rescored against the earlier *unbounded* calibration, the ceiling costs predictive skill for *only* the two runaway models (TP2, TP3), the ones that had reached 34–50 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ of manufactured litter; SP1 and the three Yassos, already inside the envelope, are unchanged. Had the good fit of TP2/TP3 rested on structure, bounding a *shared* input nuisance could not have singled those two models out. The *selectivity* of the skill drop is therefore evidence that the surplus fit was bought with unphysical carbon. We frame this as *consistent with* input/forcing error, not proof: the global design cannot exclude local kinetic variation (see the excursus and the optional stratified strand). Before/after diagnostic: Appendix A, Fig. 1.

5.2.2 Historical inputs: the unrecoverable past

No forcing data pre-observation; σ_{init} stands in for an entire history, held in the slow pool that remembers it. The deepest uncertainty.

5.2.3 Belowground inputs and other ramifications

Root/rhizodeposition inputs poorly known; a further axis of the same problem.

5.3 A short excursus: why this class of models is non-identifiable

5.3.1 The structural argument

First-order kinetics (and richer topologies) let parameters interact so the data constrain net fluxes out of a pool, not the individual splits.

5.3.1.1 Sketch

For a linear pool system the observable stocks depend on the parameters only through lumped combinations; distinct parameter sets give identical predictions (parameter-space ridges). Two equations of placeholder math. Full argument in Appendix D. [placeholder]

5.3.1.2 Signature in practice

Prior-pinning, near-perfect anti-correlation ridges between same-pool fractions, low information gain (KL of posterior vs prior). The Bayesian frame is what lets us see this rather than hide it in a fitted point. Tie to Results.

5.3.1.3 Why it is not a defect to be "fixed"

It is intrinsic to the model class and the data content (few observations per plot); it bounds what calibration can ever claim.

5.3.1.4 Managing it by convention, transparently

We do not claim to resolve the rate↔fraction equifinality — we MANAGE it: where authoritative rate values exist (Yasso) we pin the chemical axis and calibrate the partitioning; where they do not (simple models) we calibrate and let the equifinality show. An explicit convention, not an identification claim — and the candour is what keeps this section credible rather than defensive. Ties back to "degrees of freedom set by external information" in Methods.

5.4 Structure is a weak lever in-sample, but it dominates the forecast

Within the calibrated window the six structures are near-interchangeable: complexity buys no skill (Results), and the excursus just showed why — the surplus degrees of freedom are non-identified, with the inputs absorbing the misfit. Extrapolated, the picture inverts. Run forward to 2084 the models agree where data constrain them and then fan out sharply — the single-pool SP1 climbs without bound ($\sim$133 tC ha^{-1}), while the stiffer Yasso structures plateau ($\sim$108). The very structural choices the data could not pin down now govern the projection. In-sample skill is blind to this: it is scored only on the current stocks, which every model reproduces about equally well.

5.4.1 A cautious read, not a ranking

Process plausibility — not fit — gives a soft steer. Soils saturate, so SP1's unbounded single-pool rise is the least physically credible trajectory, and the saturating structures sit better with what we expect of a maturing sink. But the steer is soft and we keep its counter-caveat explicit: under the fair frame Yasso carries the most free degrees of freedom, so its plateau could be input-bound humus-parking rather than a better-resolved dynamic. Present data cannot adjudicate between these readings.

5.4.2 The honest output is the spread

Because the divergence is real, decision-relevant, and unfalsifiable from present data, the defensible product is not a chosen model but the *ensemble spread* — reported, not hidden. This is the deepest reason the national maps carry a between-model uncertainty panel beside the mean: the spatial SD is the forecast disagreement made geographic. And the projection a greenhouse-gas inventory most needs — multi-decade SOC change — is precisely the quantity in-sample skill cannot validate, which is why the honest hand-off to policy is the spread, not a single number. Narrowing that spread will not come from more SOC campaigns — two or three stocks per plot cannot arbitrate multi-decade dynamics — but from evidence that constrains the *trajectory*: long-term chronosequences and re-measured permanent plots, soil-warming and litter-manipulation experiments, and process-based limits on how much carbon a maturing boreal soil can hold. Until such constraints exist, the divergence is a genuine, reportable uncertainty, not a modelling failure.

5.5 The over-prediction is a property of the likelihood

Slight, systematic over-prediction follows from the multiplicative-normal error model, not from biogeochemistry. Only the bias is likelihood-sensitive; rankings are not.

5.6 Limitations

5.6.1 The coarse pre-observation phase

Linear litter interpolation under constant climate; high, deliberate uncertainty.

5.6.2 The flat-vs-rising litter tension

Plot-level inputs vs. national reconstructions; present data cannot settle it. State openly.

5.7 A managed landscape out of equilibrium: Finnish forest history

The concrete history behind the violated equilibrium assumption: why Finnish forest soils were accumulating, not at steady state, across the observation period. Pays off the Intro's general statement.

5.7.1 From exploitation to managed growth

A sketch of the management eras — heavy historical use, then regeneration and intensified management, rising growing stock through the 20th century. [placeholder — sources to add]

5.7.2 The imprint on litter inputs and stocks

Rising biomass and shifting management make the input history non-stationary — precisely the signal an equilibrium start erases. Connect to the σ_{input} reveal and the flat-vs-rising litter tension.

5.7.3 Why this is the rule, not a Finnish peculiarity

Most managed and recovering landscapes carry comparable histories; the lesson generalizes. Tie back to the Intro's general statement.

5.8 Outlook: toward a historical perspective on inputs

The forward call. The real next step is reconstructing historical (and belowground) input regimes.

5.8.1 From diagnosis to method (now folded into M&M)

The bound itself has moved to Methods; the remaining frontier is the input *history* (temporal) and belowground inputs. See Appendix A.

5.8.2 An interdisciplinary, source-diverse effort

Reconstruction needs evidence beyond the usual datasets — grey literature and non-traditional historical sources. Frame as a research program.

5.8.3 Initialization as a calibratable, evidence-based step

Replace the equilibrium default with priors grounded in reconstructed history, calibrated under uncertainty. The thread closes here: better history → sharper prior → a start that is assumed less and inferred more. The generalizable recommendation.

5.9 [Optional strand] Locally stratified calibration (Yasso only)

A droppable extension — include only if data + time allow; the storyline stands without it. It is the SPATIAL counterpart of the historical-reconstruction prong: both add structure to the *forcing*, never to the kinetics.

5.9.1 Scope — the operational model only

Stratify Yasso (the operational model); SP1/TP2/TP3 stay global robustness anchors. Consistent with the rate convention: Yasso already fixes its published rates, so its free knobs (inputs + fractions) are the ones available to stratify. Triggered only "if complexity emerges as needed", not by default.

5.9.2 What to stratify — inputs and the humus-formation fraction

The physically-bounded input σ and the humus-formation fraction (the stabilisation control) — NOT the rates (fixed) nor the inter-pool splits (non-identified). Both carry external, class-varying referents; humus formation is a near-net stabilisation flux the total stock genuinely constrains, so of the fractions it is the one worth stratifying.

5.9.3 RF-guided selection, holdout-evaluated

Use the residual random forest to CHOOSE the stratifying classes (its OOB R^2 is the ceiling of covariate-recoverable local structure); then measure the stratified model's gain on the HELD-OUT plots — non-negotiable, or selecting and scoring on the same data makes the improvement circular. Report "RF says up to $X\%$ is recoverable; the stratified mechanistic model recovers $Y\%$ on holdout."

5.9.4 Hierarchical, predictive-robustness framing

Partial pooling: data-poor strata shrink to the global value (no minimum-N problem); the between-stratum variance is itself a result. Headline = improved, more ROBUST posterior predictions even where stratum parameters stay weakly identified — identifiability and predictive skill are distinct, and robustness is what the GHG-inventory use needs.

5.9.5 Equifinality as a measured quantity

Bounded inputs vs. free humus formation compete for the local variance; the physical bound PARTIALLY breaks their equifinality (one competitor has walls, the other does not), so the learned variances softly attribute local variance to input vs. stabilisation origin — cross-checked by the RF driver identity (productivity proxies → input; soil-chemistry proxies → stabilisation). Frame as "consistent with", not proof. Expect modest holdout gains against a low predictability ceiling — modest-but-robust-and-honest beats an overclaimed jump.

6 Conclusions

6.1 The experiment succeeded

Transient, calibrated initialization let a model ladder reproduce the observed accumulation and its incipient saturation — the dynamic an equilibrium start forecloses. State the proof of concept plainly.

6.2 The Bayesian approach made it possible

Treating the historical start as a calibrated parameter, with uncertainty propagated into state and dynamics, is what turned the idea into a working implementation. Credit the machinery without overclaiming novelty.

6.3 Refinement needs historical data

The approach can go much further once the pre-observation history is reconstructed from data rather than left to calibration — the open frontier, and the natural successor study. Close here.

Part V
Appendices

A Physically-bounded litter-input priors

A.0.0.1 Problem.

The input multiplier σ_{input} scales the litter forcing every model integrates. Left as a weak, scale-free prior it lets the likelihood drag two models (TP2, TP3) to effective litter fluxes of 34–50 $\text{tC ha}^{-1}\,\text{yr}^{-1}$ — several times the net primary production of any boreal forest. The multiplier's distance from unity is therefore a *structural-adequacy gauge*, interpretable only once the prior encodes a physical ceiling.

A.0.0.2 Physical envelope.

The effective flux is $\sigma_{\text{input}} \times J$, with J the shared tree-litter object (median ≈ 2.8 $\text{tC ha}^{-1}\,\text{yr}^{-1}$). We bound it to the boreal total-NPP envelope of Gower et al. (2001), [0.05, 8.7] $\text{tC ha}^{-1}\,\text{yr}^{-1}$, homogeneous across all six models and applied to *both* ends of the pre-run: the contemporary flux $F_{\text{now}} = \sigma_{\text{input}}J$ and the 1917 historical flux $F_{1917} = \sigma_{\text{init}}\sigma_{\text{input}}J$. The ceiling (8.7, the Gower maximum) is the binding limit; the floor is relaxed below the Gower minimum to a small, strictly-positive, non-binding guardrail — kept > 0 so that zero flux (hence zero predicted SOC and a $-\infty$ likelihood) cannot occur, and set low to pre-position for a future stratified σ. Because σ_{init} has no independent physical referent — it is a *ratio* of two fluxes — it is not bounded on its own; instead F_{1917} is bounded to the same envelope and $\sigma_{\text{init}} = F_{1917}/F_{\text{now}}$ is recovered. The multipliers are scalar (network-average), so local disturbance (clear-cuts) lives in the per-plot J, not in the bound.

A.0.0.3 Mechanism: bounded re-parameterisation.

A hard truncation would mix poorly precisely where it bites, since the two binding models sit against the ceiling. Instead we sample an unconstrained coordinate z and map it through a scaled logistic onto the window $[a, b]$,

$$p = g(z) = a + (b - a)\,\sigma(z), \qquad \sigma(z) = \tfrac{1}{1+e^{-z}}, \qquad z = g^{-1}(p) = \log \tfrac{p-a}{b-p},$$

so every proposal is feasible by construction and the boundary is approached asymptotically, never struck. The change of variables contributes a Jacobian term to the log-prior,

$$\log\left|g'(z)\right| = \log(p - a) + \log(b - p) - \log(b - a),$$

which keeps the physical-space prior the one intended. The two fluxes are two such coordinates; the multipliers are recovered as $\sigma_{\text{input}} = F_{\text{now}}/\bar{J}$ and $\sigma_{\text{init}} = F_{1917}/F_{\text{now}}$, with $\bar{J}$ a fixed mean-litter units bridge. Implemented as one new transform type in the calibration engine, homogeneous across the six prior specifications.

A.0.0.4 Consequence.

The escape hatch closes: misfit is forced onto the (otherwise prior-pinned) kinetics or into a visible R^2 hit, exposing structural inadequacy instead of hiding it in manufactured carbon.

A.0.0.5 Before/after diagnostic.

The main text reports only the final, flux-bounded calibration. For completeness, Figure 1 contrasts it against the earlier *unbounded* homogenized run (the calibration that motivated the bound). Without the physical ceiling, TP2 and TP3 reach median effective fluxes of 34 and 50 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ — 13–20× boreal NPP, physically impossible — while SP1, Yasso07/15/20 already sit inside the envelope. Under the bound all six collapse into [0.9, 6.3], and, tellingly, *only* the two runaway models lose predictive skill: confirmation that they had been buying fit by manufacturing carbon, not by resolving kinetics. The unbounded run is retained only as this diagnostic (git-tagged, not a maintained parallel pipeline); it is not part of the reported results.

Full derivation, per-model diagnostics, literature envelope and references: standalone note `sigma_input_physical_bounds_note`*.*

B Prior homogenization: the tier scheme

[PLACEHOLDER.] The parameter-class × prior-criteria table (centre source, width source, transform/constraint, cross-model consistency) for the three tiers: climate & size, transfer fractions, auxiliary uncertainty parameters. Homogeneous in construction, scale and degrees of freedom — not in raw numbers.

B.1 Tier 1 — climate and size parameters

[placeholder table]

B.2 Tier 2 — transfer fractions

[placeholder table]

B.3 Tier 3 — auxiliary uncertainty parameters

[placeholder; σ_{init}, σ_{input} — see Appendix A for the physical bound.]

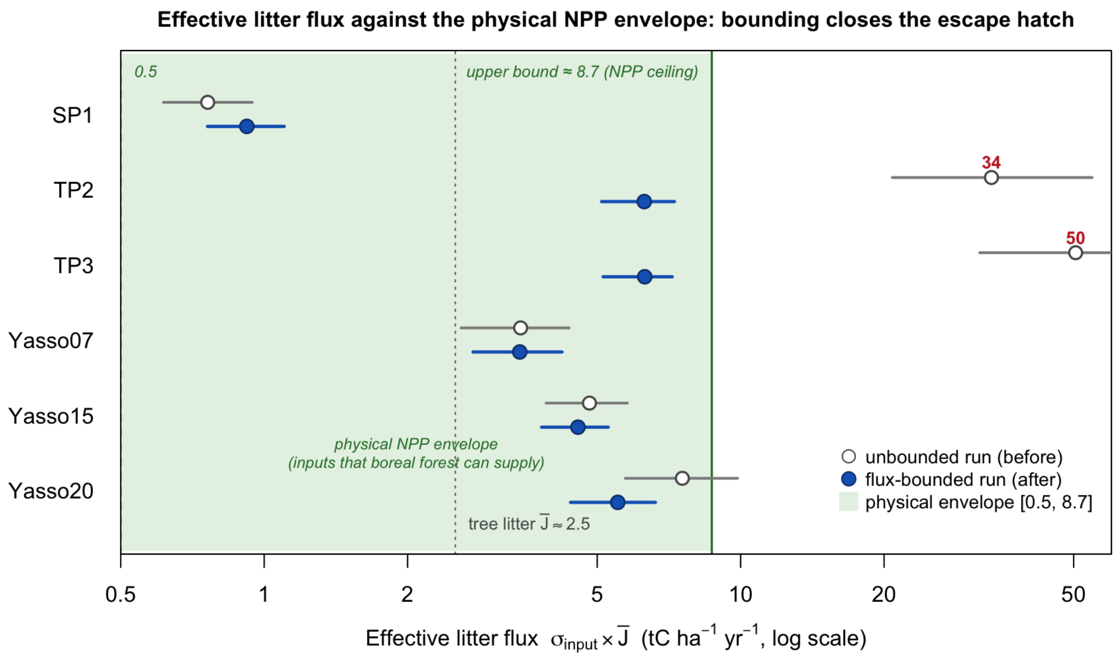

Figure 1: Effective litter flux $\sigma_{\text{input}} \times \bar{J}$ per model, on a log scale, against the physical NPP envelope [0.5, 8.7] $\text{tC}\,\text{ha}^{-1}\,\text{yr}^{-1}$ (green). Grey open points = the unbounded homogenized run (before); blue filled = the flux-bounded run (after); points are posterior medians, whiskers the 95% interval. TP2/TP3 leave the envelope entirely when unbounded (34/50) and return inside it under the bound.

C Transient initialization: the pre-run

[PLACEHOLDER.] The 1917→1985 pre-run: analytical steady state in the past, run transiently forward to t_0. Litter interpolation, constant-ξ climate during the pre-run, the fast/slow relaxation argument, and the σ_{init} anchor.

C.1 Endpoint construction and interpolation

[placeholder equations]

C.2 Fast/slow relaxation and residual memory

[placeholder]

D Non-identifiability of the linear pool class

[PLACEHOLDER.] Why the data constrain net fluxes out of a pool, not the individual splits: lumped combinations of parameters, parameter-space ridges, and the practical signature (prior-pinning, same-pool anti-correlation, low KL of posterior vs prior).

D.1 The structural argument

[placeholder equations]

D.2 Diagnostic signature

[placeholder]

E Exact integration of the pool systems

[PLACEHOLDER.] All six models integrate their linear pool system exactly: closed-form per-step (SP1, TP2), matrix exponential of the cascade generator (TP3), Fortran matrix exponential (Yasso07/15/20). Contrast with the discarded explicit-Euler path and why numerical ringing was an artefact, not structure.

E.1 Closed-form and matrix-exponential solvers

[placeholder equations]

F Reconciling the SOC-change diagnostics

Two diagnostics of the same six calibrated trajectories can appear to tell different stories, and we reconcile them here explicitly. An earlier version of the increment figure plotted the *cumulative-average* accumulation rate,

$$\bar{r}(t) \;=\; \frac{\mathrm{SOC}(t) - \mathrm{SOC}(t_0)}{t - t_0}, \qquad t_0 = 1985,$$

whose curves *decline* over time for every model. Read too quickly, a falling curve suggests SOC is decreasing — which contradicts the absolute-stock figures (the mean-SOC trajectory and the initialization figure), where every model's mean SOC *rises* toward saturation. There is no contradiction: the two are the same data under different transforms.

Figure 2 shows the identical six trajectories through three lenses. Panel (a) is the absolute stock; it rises and bends toward saturation. Panel (b) is $\bar{r}(t)$ — the earlier figure's quantity; it declines because the *time-averaged* rate of a decelerating-but-still-rising stock necessarily falls as the saturating tail is averaged in, not because any carbon is lost. Panel (c) is the instantaneous increment $\mathrm{d\,SOC}/\mathrm{d}t$; it too declines, dipping slightly negative for the multi-pool models near saturation — the derivative view of the same bend. All three panels are mutually consistent: (b)'s decline is (a)'s curvature and (c)'s deceleration.

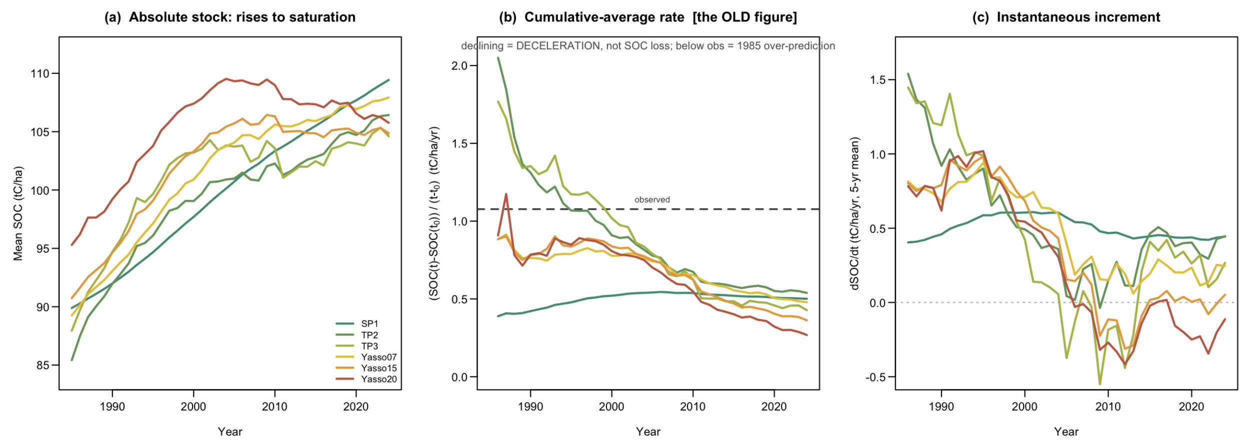

Figure 2: The same six calibrated trajectories under three transforms. (a) absolute mean SOC; (b) the cumulative-average rate $(\mathrm{SOC}(t) - \mathrm{SOC}(t_0))/(t - t_0)$ used in the earlier increment figure; (c) the instantaneous annual increment (5-year mean). The decline in (b) is the deceleration of a still-rising stock, not a loss of carbon.

Panel (b) also carries a substantive reading that the absolute-stock view makes less obvious: every model's cumulative-average rate settles *below* the observed reference (dashed). Because the models over-predict the 1985 state by $\sim 26\,\mathrm{tC\,ha^{-1}}$ (the main-text initialization figure), their accumulation *since* 1985 is smaller than the observed $63 \to 105$ climb, so their time-averaged rate cannot reach the observed value. The gap between the model curves and the dashed line in panel (b) is thus a second view of the same initialization over-prediction, not an independent discrepancy.

Notes for collapsing (remove before submission)

- **Parts** = OCAR acts; will dissolve into standard Intro/Methods/Results/Discussion at cleanup.
- Many heading levels are intentional scaffolding; merge aggressively once the story is fixed.
- Keep the protagonist (initialization problem) constant; do not let σ_{input} or non-identifiability take over the spine.